Fake Engagement Detection in Social Media — Model Explanation Document

Problem Understanding

Social media platforms face rising levels of fake engagement generated by automated bots, purchased likes, follower farms, and coordinated engagement groups. These artificial patterns distort the genuine visibility and influence of posts and users. Traditional spam filters detect obvious bots but fail to identify subtle or coordinated behaviour, especially when engagement appears real on the surface.

To ensure platform integrity, it is necessary to detect behavioural irregularities that distinguish authentic engagement from fake, artificially boosted, or automated interactions. This project aims to model user behaviour patterns and highlight suspicious activities based on timing, burst patterns, network structures, and linguistic characteristics of interactions.

Data Assumptions / Dataset Description

Since real social media behaviour logs are private, this project uses a synthetic behavioural dataset that simulates realistic engagement patterns of both:

- Genuine users
- Fake/bot/coordinated accounts

The dataset contains 15,000 users, each represented by behavioural, temporal, network, and linguistic indicators.

Dataset Features

A. Temporal Behaviour Features

Feature	Description
avg_engagement_interval	Average time gap between two actions (likes/comments)
timing_regularization	Stability of engagement timing (0–1)
nighttime_activity_ratio	Fraction of engagement happening between 1 AM–5 AM
burst_engagement_count	Number of sudden activity bursts

B. Network Interaction Features

Feature	Description
unique_profiles_interacted	Number of distinct accounts interacted with
repeated_engagement_ratio	Percentage of repeated engagement on same accounts

engagement_network_density	Density of interaction graph
suspicious_group_overlap	Overlap with known bot/boosting groups

C. Linguistic Behaviour Features

Feature	Description
comment_similarity_score	Similarity of user's comments across posts (0–1)
keyword_bot_flag	Presence of repetitive or automated phrases
language_switch_frequency	Number of times language changes across comments
sentiment_pattern_stability	Consistency of tone in comments

D. Output Labels

Label	Meaning
authenticity_score	0–1 measure of organic vs fake behaviour
bot_probability	Classifier probability of user being a bot
anomaly_type	Organic / Suspicious / Coordinated Group
flag_label	0 for real, 1 for fake engagement

3. Feature Engineering Logic

The raw data is extended with additional engineered features to capture deeper behavioural irregularities.

1. Burst Ratio

Measures sudden engagement spikes in a short time window.

burst_ratio = burst_engagement_count / total_engagements

2. Interaction Diversity Index

Captures how many unique profiles the user interacts with.

interaction_diversity = unique_profiles_interacted / max_unique_profiles

3. Repetition Score

Indicates repetitive commenting or automated-like patterns.

repetition_score = comment_similarity_score * repeated_engagement_ratio

4. Behaviour Stability Index

Aggregates multiple consistency measures.

behaviour_stability = 0.4 * timing_regularization
+ 0.3 * sentiment_pattern_stability
+ 0.3 * (1 - burst_ratio)

5. Composite Authenticity Score

Final interpretive score for organic behaviour.

$$\text{authenticity_score} = 0.5 * \text{behaviour_stability} + 0.3 * \text{interaction_diversity} + 0.2 * (1 - \text{keyword_bot_flag})$$

4. Model Selection and Reasoning

Different components of fake engagement require different ML models.

A. Bot Probability Prediction (Binary Classification)

Model: XGBoost Classifier
Reason:

- Handles noisy behavioural signals
- Works well for imbalanced bot datasets
- Provides probability outputs
- Captures nonlinear interaction patterns

B. Authenticity Score (Regression)

Model: Random Forest Regressor
Reason:

- Effective for behavioural prediction
- Works well with diverse feature types
- Produces interpretable feature importances

C. Behavioural Anomaly Detection

Model: Isolation Forest / LOF
Reason:

- Good for detecting rare anomalous patterns
- Identifies extreme behaviours without labels

D. Coordinated Group Detection

Model: Community Detection (Graph-Based Clustering)
Technique: Louvain Modularity / Spectral Clustering
Reason:

- Identifies suspicious engagement rings
- Detects behaviour overlap between fake accounts

Evaluation Metrics

A. For Classification (Bot / Not Bot)

Metric	Use
Accuracy	Overall correctness

Precision	Avoid marking real users as fake
Recall	Detect maximum fake accounts
F1 Score	Balanced metric for uneven classes
ROC-AUC	Measures ranking quality

B. For Regression (Authenticity Score)

Metric	Use
MAE	Average prediction error
RMSE	Penalizes large errors
R^2 Score	Measures predictive strength

C. For Anomaly Detection

Metric	Use
Anomaly Score Distribution	Detect outliers
Isolation Depth	Strength of anomaly
Cluster Compactness	Coordinated bot group detection

Behavioural Insights Derived

- Extremely short engagement intervals indicate automated behaviour.
- High repetitive comment similarity is strongly tied to bot-like patterns.
- Users interacting with a very limited set of profiles show coordinated group behaviour.
- Sudden engagement bursts within seconds are clear artificial indicators.
- High night-time activity ratio correlates with bot clusters.
- Organic users show diverse connections and varied linguistic patterns.
- Comment similarity + repeated engagement ratio = strongest fake predictor.